

RESTAURANT DATA ANALYSIS USING PYTHON AND POWER BI

1. INTRODUCTION

Data analysis plays a crucial role in understanding customer preferences and business trends. In the food and restaurant industry, analyzing data can help identify popular cuisines, customer satisfaction levels, and service trends.

This project focuses on analyzing restaurant data using Python for data preprocessing and Power BI for interactive visualization. The objective is to extract meaningful insights related to cuisines, ratings, pricing, and customer behavior.

2. OBJECTIVES

The main objectives of this project are:

- To identify the most popular cuisines
 - To analyze city-wise restaurant ratings
 - To study the distribution of restaurants across price ranges
 - To examine online delivery availability
 - To analyze the relationship between votes and ratings
 - To identify restaurant chains and their performance
 - To explore geographic distribution of restaurants
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3. DATASET DESCRIPTION

The dataset contains information about various restaurants, including their location, ratings, cuisines, and services offered.

Key Attributes:

- Restaurant Name
- City
- Cuisines
- Aggregate Rating
- Votes
- Price Range
- Has Online Delivery

- Latitude and Longitude

This dataset helps in understanding patterns in restaurant operations and customer preferences.

4. TOOLS AND TECHNOLOGIES USED

The following tools were used in this project:

- Python – for data cleaning and analysis
 - Power BI – for dashboard creation and visualization
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5. DATA PREPROCESSING

Data preprocessing was performed using Python to ensure data quality and consistency.

The following steps were carried out:

- Removal of missing values
- Cleaning of column names
- Handling inconsistent data entries
- Conversion of data types

Sample Code:

```
import pandas as pd

df = pd.read_csv("Dataset.csv")
df = df.dropna()
df.columns = df.columns.str.strip()
```

6. DATA ANALYSIS AND VISUALIZATION

6.1 Top Cuisines Analysis

Title: *Top Cuisines by Number of Restaurants*

This analysis identifies the most popular cuisines based on the number of restaurants offering them. It helps in understanding customer food preferences and market demand.

6.2 City-wise Rating Analysis

Title: *Top Cities by Average Restaurant Rating*

This visualization compares average ratings across different cities to identify locations with higher customer satisfaction.

6.3 Price Range Distribution**Title:** *Distribution of Restaurants by Price Range*

This chart shows how restaurants are distributed across different price categories, highlighting affordability trends.

6.4 Online Delivery Analysis**Title:** *Online Delivery Availability of Restaurants*

This analysis represents the proportion of restaurants offering online delivery services, reflecting the importance of digital convenience.

6.5 Votes vs Rating Analysis**Title:** *Relationship Between Votes and Restaurant Ratings*

This scatter plot examines the relationship between customer engagement (votes) and restaurant ratings.

6.6 Cuisine Combination Analysis**Titles:**

- *Top Cuisine Combinations by Number of Restaurants*
- *Average Ratings by Cuisine Combination*

This analysis explores commonly paired cuisines and their performance in terms of ratings.

6.7 Geographic Analysis**Title:** *Geographical Distribution of Restaurants*

This visualization maps restaurant locations using latitude and longitude to identify clusters and high-density areas.

6.8 Restaurant Chains Analysis

Titles:

- *Top Restaurant Chains by Number of Outlets*
- *Average Ratings of Restaurant Chains*

This analysis identifies popular restaurant chains and evaluates their performance across multiple outlets.

7. KEY INSIGHTS

- A few cuisines dominate the restaurant market
 - Most restaurants fall within the mid-price range
 - Online delivery services are widely available
 - Certain cities have higher average ratings than others
 - The relationship between votes and ratings is relatively weak
 - Some restaurant chains have a strong market presence
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8. CONCLUSION

The project successfully analyzed restaurant data to uncover valuable insights into customer preferences, pricing trends, and service availability. The combination of Python and Power BI enabled effective data cleaning, analysis, and visualization.

These insights can help businesses improve their strategies and enhance customer satisfaction.

9. LIMITATIONS

- The dataset may contain missing or inconsistent data
 - Analysis is limited to available attributes
 - No real-time data updates were considered
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10. FUTURE SCOPE

- Implementation of machine learning models for prediction
 - Integration of real-time data sources
 - Development of recommendation systems for customers
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11. REFERENCES

- Dataset provided for analysis
- Python documentation
- Power BI official documentation